MARKSCHEME CLASS TST AAHL 27 OCT 25

log$x$+1 y = 2

log$y$+1 x = $\frac{1}{4}$

so (x + 1)2 = y A1

$(y+1)^{\frac{1}{4}}$ = x A1

EITHER

x4 – 1 = (x + 1)2 M1

x = –1, not possible R1

x = 1.70, y = 7.27 A1A1

OR

$(x^2+2x+2)^{\frac{1}{4}}$ – x = 0 M1

attempt to solve or graph of LHS M1

x = 1.70, y = 7.27 A1A1

[6]

u4 = u1 + 3d = 7, u9 = u1 + 8d = 22 A1A1

Note: 5d = 15 gains both above marks

u1 = –2, d = 3 A1

Sn = $\frac{n}{2}$ (–4 + (n – 1)3) > 10000 M1

n = 83 A1

[5]

3.

(a) 18n – 10 (or equivalent) A1

(b) $\sum_{1}^{n}(18r-10)$ (or equivalent) A1

(c) by use of GDC or algebraic summation or sum of an AP (M1)

$\sum_{1}^{15}(18r-10)$ = 2010 A1

[4]

4.

METHOD 1

(a) $u_n = S_n - S_{n-1}$ (M1)

$= \frac{7^n - a^n}{7^n} - \frac{7^{n-1} - a^{n-1}}{7^{n-1}}$ A1

(b) **EITHER**

$u_1 = 1 - \frac{a}{7}$ A1

$u_2 = 1 - \frac{a^2}{7^2} - \left(1 - \frac{a}{7}\right)$ M1

$= \frac{a}{7}\left(1 - \frac{a}{7}\right)$ A1

common ratio $= \frac{a}{7}$ A1

OR

$u_n = 1 - \left(\frac{a}{7}\right)^n - 1 + \left(\frac{a}{7}\right)^{n-1}$ M1

$= \left(\frac{a}{7}\right)^{n-1}\left(1 - \frac{a}{7}\right)$

$= \frac{7-a}{7}\left(\frac{a}{7}\right)^{n-1}$ A1

$u_1 = \frac{7-a}{7}$, common ratio $= \frac{a}{7}$ A1A1

(c) (i) $0 < a < 7$ (accept $a < 7$) A1

(ii) 1 A1

METHOD 2

(a) $u_n = br_{n-1} = \left(\frac{7-a}{7}\right)\left(\frac{a}{7}\right)^{n-1}$ A1A1

(b) for a GP with first term b and common ratio r

$S_n = \frac{b(1-r^n)}{1-r} = \left(\frac{b}{1-r}\right) - \left(\frac{b}{1-r}\right)r^n$ M1

as $S_n = \frac{7^n - a^n}{7^n} = 1 - \left(\frac{a}{7}\right)^n$

comparing both expressions M1

$\frac{b}{1-r}$ = 1 and r = $\frac{a}{7}$

b = 1 $-\frac{a}{7}=\frac{7-a}{7}$

$u1$ = b = $\frac{7-a}{7}$, common ratio = r = $\frac{a}{7}$ A1A1

Note: Award method marks if the expressions for b and r are deduced in part (a).

(c) (i) $0 < a < 7$ (accept $a < 7$) A1

(ii) 1 A1

[8]

5.

$22x+2 - 10 \times 2x + 4 = 0$
$y = 2x$
$4y2 - 10y + 4 = 0$ M1A1
$2y2 - 5y + 2 = 0$

By factorisation or using the quadratic formula (M1)

$y = \frac{1}{2}$ $y = 2$ A1

$2x = \frac{1}{2}$ $2x = 2$
$x = -1$ $x = 1$ A1A1

[6]

6.

correct application of $\log_a xy = \log_a x + \log_a y$ **OR** $\log_a x^m = m \log_a x$ *(M1)*

correct expression in terms of $\log_{10} 2$ **AND** $\log_{10} 3$ (arguments must be 2 and 3) *(A1)*

$3 \log_{10} 2 + \log_{10} 3$ **OR** $\log_{10} 2 + \log_{10} 2 + \log_{10} 2 + \log_{10} 3$

$3p + q$ *A1*

[3 marks]

(b) Find an expression for $\log_3 8$ in terms of p and q.

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$(\log_3 8 =) \frac{\log_{10} 8}{\log_{10} 3} \left(= \frac{3 \log_{10} 2}{\log_{10} 3}\right)$ *(A1)*

$= \frac{3p}{q}$ *A1*

[2 marks]

7.

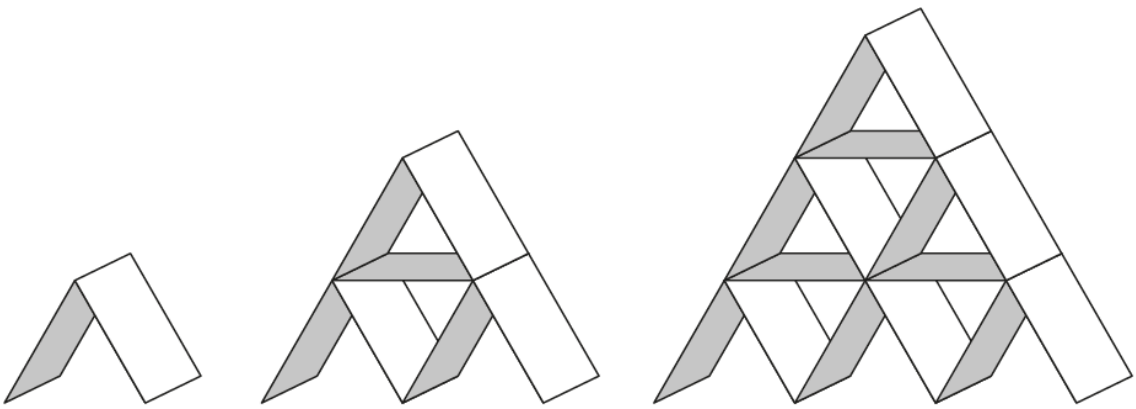
8.

[Maximum mark: 16]

Rectangular playing cards are stacked in the shape of a pyramid with rows, where .

Some cards are placed horizontally and some cards are stacked at an angle of to the horizontal.

The following diagrams represent pyramid stacks for , and .



$n = 1$ $n = 2$ $n = 3$

attempt to add cards onto a stack with rows **OR** attempt to consider all rows ***(M1)***

valid diagram with rows **OR** **OR**

Markscheme

15 *A1*

[1 mark]

attempt to add 11 cards onto a stack with 3 rows **OR** attempt to consider all 4 rows *(M1)*

valid diagram with 4 rows **OR** $t_4 = 15 + 11$ **OR** $t_4 = 2 + 5 + 8 + 11$

$= 26$ *A1*

[2 marks]

recognition that t_n is a sum of an arithmetic sequence *(M1)*

$t_n = 2 + 5 + 8 + 11 + \ldots$

attempt to use formula for the sum of n terms of an arithmetic sequence *M1*

$t_n = \frac{n}{2}(2(2) + 3(n-1))$ *A1*

$t_n = \frac{n}{2}(3n+1)$ *AG*

METHOD 2

attempt to split t_n into the total number of stacked and horizontal cards *(M1)*

stacked $2 + 4 + 6 + \ldots = \frac{n}{2}(4 + 2(n-1))\left(= \frac{n}{2}(2n+2)\right)$ *A1*

horizontal $0 + 1 + 2 + \ldots = \frac{n}{2}(0 + 1(n-1))\left(= \frac{n}{2}(n-1)\right)$ *A1*

$t_n = \frac{n}{2}(4 + 2(n-1)) + \frac{n}{2}(0 + 1(n-1))\left(= \frac{n}{2}(2n+2) + \frac{n}{2}(n-1)\right)$

$t_n = \frac{n}{2}(3n+1)$ *AG*

recognition that a stack with n rows is made up of complete triangles with the bottom row of horizontal cards removed and that the numbers of complete triangle cards form an arithmetic sequence *(M1)*

$t_n = (3 + 6 + 9 + 12 + \ldots + 3n) - n$ **OR** $t_n = 3(1 + 2 + 3 + 4 + \ldots + n) - n$

attempt to use formula for the sum of n terms of an arithmetic sequence *M1*

$t_n = \frac{n}{2}(2(3) + 3(n-1)) - n$ **OR** $t_n = 3 \times \frac{n}{2}(1+n) - n$ *A1*

$t_n = \frac{n}{2}(3n+1)$ *AG*

[3 marks]

attempt to solve $\frac{n(3n+1)}{2} \leq 14(52)\ (= 728)$ *(M1)*

Note: Accept an attempt to solve an equation for *(M1)*.

$21.8642\ldots$ **OR** $n = 21,\ t_n = 672$ and $n = 22,\ t_n = 737$ *(A1)*

max number of rows is 21 *A1*

[3 marks]

attempt to solve by listing at least six values of t_n *(M1)*

$2,\ 7,\ 15,\ 26,\ 40,\ 57\ldots$

OR

recognition that $\frac{\frac{1}{2}n(3n+1)}{52}$ must be an integer *(M1)*

$\frac{1}{2}n(3n+1) = 52k$ (where k is an integer)

THEN

min number of rows is 13 *A1*

Note: Award *(M1)A0* for an answer of 5 packs.
Award *M0A0* for any answer resulting from solving $\frac{1}{2}n(3n+1) = 52$.

[2 marks]

METHOD 2

$\mathrm{PV} = \pm 30000$

$\mathrm{FV} = \mp 5906.23213\ldots$

$\mathrm{I\%} = -15\%$

$\mathrm{P\ /\ Y} = 1$

$\mathrm{C\ /\ Y} = 1$

$\mathrm{N} = 10$ *(M1)*

€ 5906.23 *A1*

[2 marks]

METHOD 1

$r = c - i\,(= 1.5 - 0.8)$

$r = 0.7\,(\%)$ *(A1)*

attempt to substitute their values into inequality *(M1)*

$50000\left(1 + \frac{0.7}{100}\right)^n > 55000$ (accept equality)

correct critical value **OR** at least one correct crossover value *(A1)*

$n = 13.6633\ldots$ **OR** $n = 13,$ € 54746.09 **OR** $n = 14,$ € 55129.31

14 (months) *A1*